Undergraduate Seminar
Introduction to Commutative Algebra

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Preface

Why this note exists?

Moving to Japan with no prior knowledge of the Japanese language was one of the most challenging phases of my life. I still remember sitting through Foundational Mathematics I and II (数学の基礎 I, II) lectures and not understanding a single word. Because of the language barrier, those lectures were an incredibly stressful and difficult experience.

Reflecting on that time, I realize how hard it is to study a new subject while struggling with a new language. **I wrote these notes to act as a bridge, helping the attendees understanding the lecture material in English**. While there may be a little bit of Japanese used, the presentation will be delivered primarily in English.

To the attendees

Thank you so much for attending. **You are welcome to ask questions or offer comments in Japanese**[1].

Technical details

This seminar follows Atiyah Macdonald's Introduction to Commutative Algebra. Our goal is to study these topics in depth and fill in the details often omitted by the book.

Commutative Algebra is famously **dry**, and the authors do not always provide the underlying motivation for certain concepts or results. To counter this, **I tried to include as many examples as possible**.

I have also included Japanese translations for some terms as a footnote.

Acknowledgements

Finally, I wish to express my gratitude to Professor 佐藤 for the opportunity to conduct this seminar in English and for his valuable guidance. My thanks also go to my colleagues 名取, 英, and 松永, and to everyone else in attendance.

Romano Emmanuelle

[1] However, since I am still learning japanese, I might get a confused, so please do not hesitate to switch to English if you would like!

序文

なぜこのノートを作成したのか？

日本語が全くわからない状態で日本に来たことは、人生で一番大変な時期でした。特に数学の基礎 I・II の講義で、単語が一つもわからなかった時のことは今でも記憶に残ります。言葉の壁のせいで、あの頃は本当に苦労しました。

その経験から、新しいことを慣れない言語で学ぶ難しさがよくわかりました。そこで、**皆様が英語で講義を理解するための「架け橋」になるように、このノートを作りました**。発表では少し日本語も使いますが、殆ど英語で話す予定です。

聴講者の皆様へ

今日は来てくれてありがとうございます。質問やコメントは日本語でも大丈夫です[1]。

テクニカルな内容について

このゼミは、アティヤマクドナルドの可換代数入門に基づいています。私は、本の中で省略されている細かい部分を補いながら、トピックをより深く勉強することに重点を置いています。

この分野の内容は非常に**「ドライ（抽象的？）」**であることで知られていて、著者も概念や結果についての具体的な動機（モチベーション？）をあまり書いていません。そのため、**私はできるだけ多くの具体例を含めるように努めました**。

理解を助けるために、いくつかの単語には日本語の訳も付けています。

謝辞

最後に、英語でこのゼミをする機会をくださり、熱心に指導してくださった佐藤先生に心から感謝いたします。また、いつも助けてくれるメンバーの名取様、英様、松永様、そして今日ここに来てくださった皆様、本当にありがとうございました。

ロマノエマニュエル

[1] ただし、まだ日本語に苦戦しているので、英語で話してくれてむしろ大歓迎です！

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April, 22 2026

§1.1 Nilradical and Jacobson Radical (1.5)

let us recall[1] the definition of nilpotent:

Remark — An element[a] $a \in A$ is nilpotent[b] if $a^n = 0$ for some $n > 0$.

[a] 元
[b] ベキ零元

Example (Sanity Check)

1. For $A = \mathbb{Z}$, $x = 2$ is **not** nilpotent. (There is no n such that $2^n = 0$)
2. For $A = \mathbb{Z}/8\mathbb{Z}$, $x = 2$ is nilpotent because $2^3 = 0$.
3. For $A = \mathbb{Z}/6\mathbb{Z}$, $x = 0$ is the only nilpotent element.

Let A be a ring. We denote $\mathfrak{R}$ as the set[2] of all nilpotent elements in A[3]. We write $\mathfrak{R}$ as the following:

$$\mathfrak{R} := \{a \in A \mid \text{there exists } n > 0 \text{ such that } a^n = 0\}.$$

Proposition 1.1 (1.7)

Let $A \neq 0$ be a ring[a]. Then the following hold[b]:

i) $\mathfrak{R}$ is an ideal of A.

ii) $A/\mathfrak{R}$ has no nilpotent element $\neq 0$.

[a] 環
[b] 次が成り立つ

The ideal $\mathfrak{R}$ (Often denoted by $\mathrm{Nil}(A)$ or $\sqrt{(0)}$ in the literature) is called the **nilradical**[4] of A.

Proof. Our goal is to prove $x, y \in \mathfrak{R} \Rightarrow x + y \in \mathfrak{R}$ and $ax \in \mathfrak{R}$ for every $a \in A$, $x, y \in \mathfrak{R}$.

[1] 復習 [2] 集合 [3] ”環 A におけるすべてのベキ零元の集合” [4] ベキ零元根基

a) Suppose[5] $x, y \in \mathfrak{R}$. Then, there exists[6] m, n such that $x^m = 0$, $y^n = 0$. Now, by the binomial theorem[7], we have

$$(x+y)^{m+n-1} = \sum_{r+s=m+n-1; r,s\geq 0} \binom{m+n-1}{r} x^r y^s.$$

Notice that either $r \geq m$ or $s \geq n$ (see remark below), which results to

$$(x+y)^{m+n-1} = \underbrace{0+0+\cdots+0}_{m+n-1} = 0.$$

Therefore, there exists a constant c (in particular[8] $m+n-1$) such that $(x+y)^c = 0$. Thus $x+y \in \mathfrak{R}$.

b) Suppose $x \in \mathfrak{R}$ and $a \in A$. Then, there exists $m > 0$ such that $x^m = 0$. Therefore, $ax \in \mathfrak{R}$ because $(ax)^m = a^m x^m = 0$[9].

ii) Let $\tilde{x} \in A/\mathfrak{R}$ be a nilpotent element:

$$\tilde{x} := x + \mathfrak{R} \in A/\mathfrak{R}.$$

We will show that $\tilde{x} = 0$. Since $\tilde{x}$ nilpotent, there exists n such that $\tilde{x}^n = 0$. Then,

$$0 = \tilde{x}^n = x^n + \mathfrak{R}.$$

This implies $x^n \in \mathfrak{R}$. Consequently[10], there exists $k \in \mathfrak{R}$ such that $(x^n)^k = 0$. This means that $x \in \mathfrak{R}$. Finally, $\tilde{x}^n = x + \mathfrak{R} = 0$, concluding that $\tilde{x} = 0$. We have shown that there is **no** nilpotent element other than 0.

□

Remark — For the sake of contradiction[a] assume $r < m-1$ and $s < n-1$. Then,

$$r + s \leq (m-1) + (n-1) = m + n - 2.$$

This is a contradiction[b] because $r + s = m + n - 1$. Therefore, either $r \geq m$ or $s \geq n$ holds.

[a] FTSoC; 矛盾を導くために or 背理法で示す,
[b] 矛盾

Example

Let $A = \mathbb{Z}/12\mathbb{Z}$ be a ring, then $\mathfrak{R} = (6)$ is an ideal of A. 0 is the only nilpotent element of $A/\mathfrak{R}$.

The following proposition gives an alternative[11] definition of $\mathfrak{R}$:

[5] 仮に [6] 存在 [7] 二項定理 [8] 特に [9] This only true when A is a commutative ring! [10] よって [11] 代替

Proposition 1.2 (1.8)

The nilradical $\mathfrak{R}$ of A is the intersection[a] of all prime ideal[b] of A. That is,

$$\mathfrak{R} = \bigcap_{A \supsetneq \mathfrak{q} \text{ prime}} \mathfrak{q}.$$

[a] 交わり
[b] 素イデアル

To prove the proposition above, we need Zorn lemma:

Lemma 1.3 (Zorn's Lemma)

Let S be poset[a] that satisfies the following two properties:

1. S is nonempty[b].
2. Every chain[c] T in S has an upper bound[d] **in** S.

Then, S has atleast one maximal element.

[a] "Partially ordered set": 順序集合
[b] 空でない
[c] 鎖
[d] 上限

We omit[12] the proof of Zorn's Lemma and proceed to the proof of proposition 1.2 (1.8).

Proof. Write $\bigcap_{A \supsetneq \mathfrak{q} \text{ prime}} \mathfrak{q} =: \mathfrak{R}'$. We will show that a) $\mathfrak{R} \subseteq \mathfrak{R}'$ and b) $\mathfrak{R}' \subseteq \mathfrak{R}$.

a) For each[13] $f \in \mathfrak{R}$, there exists $n > 0$ such that $f^n = 0$. On the other hand[14], by the definition of ideal[15], for each ideal prime $\mathfrak{q}$ we have

$$\mathfrak{q} \ni 0 = f^n.$$

Therefore, $f \in \mathfrak{q}$[16]. Finally, since $\mathfrak{q}$ is arbitrary[17], we conclude that $f \in \mathfrak{R}'$.

b) Assume $f \notin \mathfrak{R}$, we will show that $f \notin \mathfrak{R}'$. By assumption, for any $n > 0$, $f^n \neq 0$. Now, we will use Zorn's Lemma. Define

$$S := \{\mathfrak{a} \subseteq A \mid \mathfrak{a} \text{ is an ideal and } f^n \notin \mathfrak{a} \text{ for each } n > 0\}.$$

1. S is nonempty because $0 \in S$.
2. Order the ideals $\mathfrak{a}$ by inclusion (say, $\mathfrak{a}_\alpha \subseteq \mathfrak{a}_\beta$ or $\mathfrak{a}_\beta \subseteq \mathfrak{a}_\alpha$ for each α, β). Then, every chain $T := (\mathfrak{a}_\alpha)_{\alpha \in \Lambda}$ in S has an upper bound[18], namely

$$\mathfrak{a} := \bigcup_{\alpha \in \Lambda} \mathfrak{a}_\alpha.$$

[12] 省略する [13] 任意の [14] 一方 [15] イデアルの定義より [16] Notice that $f^n = f \cdot f^{n-1} \in \mathfrak{q}$ implies $f \in \mathfrak{q}$ or $f^{n-1} \in \mathfrak{q}$. [17] $\mathfrak{q}$ は任意なので [18] We need to prove $\mathfrak{a} \in S$!

Then, By Zorn's Lemma, S has at least one maximal element, say $\mathfrak{p}$. Consequently, $\mathfrak{p} \in S$ and $f \notin \mathfrak{p}$.

Claim. $\mathfrak{p}$ is a prime ideal. (See remark for the proof)
The claim implies $f \notin \mathfrak{R}'$. This concludes the proof.

□

Remark (Proof of the claim) — For every $x, y \in \mathfrak{p}$, suppose $x \notin \mathfrak{p}$ and $y \notin \mathfrak{p}$. Our goal is to show $xy \notin \mathfrak{p}$.
First, notice that $(x) + \mathfrak{p}, (y) + \mathfrak{p} \supsetneq \mathfrak{p}$ because $x, y \notin \mathfrak{p}$. Furthermore[a], since $\mathfrak{p}$ is the maximal element of S, we have $\mathfrak{p} + (x), \mathfrak{p} + (y) \notin S$.
Consequently[b], there exist m and n such that $f^m \in \mathfrak{p} + (x)$ and $f^n \in \mathfrak{p} + (y)$.
Now, write

$$f^m = p_x + ax \quad p_x \in \mathfrak{p}, a \in A$$
$$f^n = p_y + bx \quad p_y \in \mathfrak{p}, b \in A.$$

Then,

$$f^{m+n} = \underbrace{p_x p_y + p_x by + p_y ax}_{\in \mathfrak{p}} + \underbrace{abxy}_{\in (xy)},$$

which results to $f^{m+n} \in \mathfrak{p} + (xy)$. Finally, since $\mathfrak{p} + (xy) \notin S$ and $\mathfrak{p} \in S$, we have $xy \notin \mathfrak{p}$.

[a] 更に
[b] その結果

Remark — The $\mathfrak{p}$ stipulated above[a] is **not** equal to $\mathfrak{R}'$!

[a] 上記に定められた

Remark (Diggresion) — ~~I think the propoerty 1. of the Zorn's Lemma is somehow redundant[a], since property 2. says, in particular that the empty chain has an upper bound in S, automatically implies that S itself is nonempty...?~~
Update: after consulting to Professor 佐藤, the condition 1. of the Zorn's Lemma is **necessary** so that condition 2. holds true.

[a] 余分? 必要以上に存在する

Example
Let $A = \mathbb{Z}/12\mathbb{Z}$, then

$$(6) = \mathfrak{R} = \bigcap_{A \supseteq \mathfrak{q} \text{ prime}} \mathfrak{q} = (2) \cap (3),$$

as desired.

We have defined **nilradical** $\mathfrak{R}$ to be the intersection of all prime ideal of a ring. Similarly, we define **Jacobson radical**[19] $\mathfrak{J}$[20] of A to be the intersection of all maximal ideals of A. It can be

[19] ジャコブソン根基
[20] The Japanese and English version of the book uses the same symbol $\mathfrak{R}$. However, I find it uncomfortable and therefore introducing $\mathfrak{J}$ as the alternative.

characterized as follows:

Proposition 1.4 (1.9)
$x \in \mathfrak{J} \iff 1 - xy$ is a unit in A for all $y \in A$.

Proof.

⇒) Suppose $x \in \mathfrak{J}$. For the sake of contradiction assume there exists $y \in A$ such that $1 - xy$ is **not** a unit in A. Then, by Corollary 1.5, $1 - xy$ belongs inside some maximal ideal, say $\mathfrak{m}$. Furthermore, $x \in \mathfrak{J} \subseteq \mathfrak{m}$. On the other hand, by the definition of ideal we have $yx \in \mathfrak{m}$. This implies $1 \in \mathfrak{m}$, a contradiction!

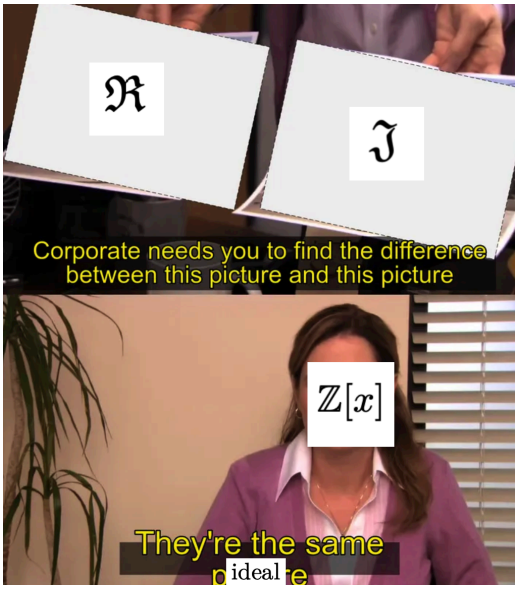
⇐) Now, suppose $1 - xy$ is a unit in A for all $y \in A$. For the sake of contradiction assume $x \notin \mathfrak{m}$. Then, we have $\mathfrak{m} + (x) = (1)$, that is, $\mathfrak{m}$ and x generate the unit ideal (1). Consequently, there exist $m \in \mathfrak{m}$ and $y \in A$ such that $m + xy = 1$. Therefore, we have $1 - xy \in \mathfrak{m}$. Hence $1 - xy$ is not a unit (by Corollary 1.5), contradicting the given assumption.

□

Remark — In general, we only have the inclusion $\mathfrak{R} \subseteq \mathfrak{J}$. Equality may be achieved by stipulating some restrictions on A[a] (e.g., assuming A is an **Artinian ring**, where the descending chain of ideals eventually stabilizes, etc).

[a] A に何らかの制限を設けることによって、$\mathfrak{R} = \mathfrak{J}$ である.

Exercise 1.5. Deduce the mathematical result from the following picture:



§1.2 Operation on Ideals (1.6)

In the previous section, we introduced[21] the nilradical and the Jacobson radical. We now turn to basic operations[22] on ideals and how they interact.

Let $\mathfrak{a}$ and $\mathfrak{b}$ be ideals of a ring A. Their **sum** is defined by the following **set**:

$$\mathfrak{a}+\mathfrak{b} := \{x+y \mid x\in\mathfrak{a},\ y\in\mathfrak{b}\}.$$

This is the **smallest ideal** of A containing both $\mathfrak{a}$ and $\mathfrak{b}$[23].

Exercise 1.6. Verify[24] that $\mathfrak{a}+\mathfrak{b}$ is an ideal of A, and that it is indeed the smallest ideal containing $\mathfrak{a}$ and $\mathfrak{b}$.

Generally, given a family[25] of ideals $(\mathfrak{a}_i)_{i\in I}$ of A, their sum is defined by

$$\sum_{i\in I}\mathfrak{a}_i := \left\{ \sum_{i\in I} x_i \;\middle|\; x_i\in\mathfrak{a}_i, \text{ and } \textbf{only finitely many } x_i\neq 0 \right\}.$$

This is again an ideal of A, and it is the smallest ideal containing all the $\mathfrak{a}_i$.

> **Remark** — What happen if you ignore[a] the
>
> "**only finitely many** $x_i \neq 0$" ?
>
> [a] 無視する

On the other hand, the **intersection**[26] of any family $(\mathfrak{a}_i)_{i\in I}$ of ideals is also an ideal of A.
Finally, the **product** of two ideals $\mathfrak{a},\mathfrak{b}\subseteq A$ is defined by

$$\mathfrak{a}\mathfrak{b} := \langle xy \mid x\in\mathfrak{a},\ y\in\mathfrak{b}\rangle,$$

that is, an ideal generated[27] by all products xy, where $x\in\mathfrak{a}, y\in\mathfrak{b}$. $\mathfrak{a}\mathfrak{b}$ is again an ideal of A.

Exercise 1.7. Why don't we define $\mathfrak{a}\mathfrak{b}$ as

$$\mathfrak{a}\mathfrak{b} := \{x+y \mid x\in\mathfrak{a}, y\in\mathfrak{b}\}\ ?$$

Similarly[28], we define the product of any **finite** family of ideals $(\mathfrak{a}_i)_{i\in I}$ as following:

$$\prod_{i\in I}\mathfrak{a}_i := \langle\ x_1\cdots x_n \mid x_i\in\mathfrak{a}_i \text{ for every } i\in I\ \rangle.$$

In particular[29], for $n>0$, the power $\mathfrak{a}^n$ is generated[30] by all finite products $x_1\cdots x_n$ with $x_i\in\mathfrak{a}$.

[21] 導入した [22] 演算 [23] $\mathfrak{a}$ と $\mathfrak{b}$ を含んでいる A の最小のイデアルである [24] 確かめよ [25] 集合族 [26] 交わる [27] 生成する [28] 同様に [29] 特に [30] 生成される

Example

1) Let $A = \mathbb{Z}$, $\mathfrak{a} = (m)$, $\mathfrak{b} = (n)$. Then, we have the following:
   i) $\mathfrak{a} + \mathfrak{b}$ is the ideal generated by the GCD[a] of m and n,
   ii) $\mathfrak{a} \cap \mathfrak{b}$ is the ideal generated by the LCM[b] of m and n,
   iii) $\mathfrak{ab} = (mn)$.

   Thus $\mathfrak{ab} = \mathfrak{a} \cap \mathfrak{b} \iff m, n$ are coprime[c].

2) Let $A = k[x_1, \ldots, x_n]$ and $\mathfrak{a} = (x_1, \ldots, x_n)$ is the ideal generated by $x_1, \ldots, x_n$. Then $\mathfrak{a}^m$ is the set of all polynomials whose terms all have degree $\geq m$.

[a] Greatest Common Divisor, 最大公約数
[b] Lowest Common Multiplier, 最小公倍数
[c] 互いに素

Remark — The three operations so far defined (sum, intersection, product) are all commutative[a] and associative[b].

[a] 可換
[b] 結合的

In $\mathbb{Z}$	In a general ring A
$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) = \mathfrak{a} \cap \mathfrak{b} + \mathfrak{a} \cap \mathfrak{c}$	$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) \supseteq \mathfrak{a} \cap \mathfrak{b} + \mathfrak{a} \cap \mathfrak{c}$
$(\mathfrak{a} + \mathfrak{b})(\mathfrak{a} \cap \mathfrak{b}) = \mathfrak{ab}$	$(\mathfrak{a} + \mathfrak{b})(\mathfrak{a} \cap \mathfrak{b}) \subseteq \mathfrak{ab}$
$\mathfrak{a} + \mathfrak{b} = (1) \Rightarrow \mathfrak{a} \cap \mathfrak{b} = \mathfrak{ab}$	
$\mathfrak{a}(\mathfrak{b} + \mathfrak{c}) = \mathfrak{ab} + \mathfrak{ac}$	

Remark — In a general ring A,

$$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) = \mathfrak{a} \cap \mathfrak{b} + \mathfrak{a} \cap \mathfrak{c}$$

if $\mathfrak{a} \supseteq \mathfrak{b}$ or $\mathfrak{a} \supseteq \mathfrak{c}$.

Two ideals $\mathfrak{a}, \mathfrak{b}$ are said to be **coprime** (or **comaximal**) if $\mathfrak{a} + \mathfrak{b} = (1)$. Thus for coprime ideals we have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{ab}$. Clearly two ideals $\mathfrak{a}, \mathfrak{b}$ are coprime if and only if there exist $x \in \mathfrak{a}$ and $y \in \mathfrak{b}$ such that $x + y = 1$.

Let $A_1, \ldots, A_n$ be rings. Their **direct product**

$$A = \prod_{i=1}^{n} A_i$$

is the set of all sequences $x = (x_1, \ldots, x_n)$ with $x_i \in A_i$ $(1 \leq i \leq n)$ and componentwise[31] addition and multiplication. A is a commutative ring with identity element $(1, 1, \ldots, 1)$. We have projections[32] $p_i : A \to A_i$ defined by $p_i(x) = x_i$; they are ring homomorphisms[33].

[31] 成分ごと [32] 射影 [33] 準同型

Let A be a ring and $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$ ideals of A. Define a homomorphism

$$\phi : A \longrightarrow \prod_{i=1}^{n} (A/\mathfrak{a}_i)$$

by $\phi(x) := (x + \mathfrak{a}_1, \ldots, x + \mathfrak{a}_n)$.

Exercise 1.8. Show that ϕ is indeed a homomorphism!

2

June, 17 2026

§2.1 Exact Sequences (2.6)

A sequence[1] of A-modules[2] and A-homomorphisms[3]

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots \tag{0}$$

is said to be **exact at** M_i[4] if

$$\operatorname{im}(f_i) = \ker(f_{i+1}).$$

The sequence is said to be **exact** if it is exact at every M_i. In particular, we shall use the following basic[5] fact:

Proposition 2.1

Let M, M', M'' be A-modules, and let $f : M' \to M$ and $g : M \to M''$ be A-module homomorphisms. Then,

$$0 \longrightarrow M' \xrightarrow{f} M \text{ is exact} \iff f \text{ is injective, and} \tag{1}$$

$$M \xrightarrow{g} M'' \longrightarrow 0 \text{ is exact} \iff g \text{ is surjective.} \tag{2}$$

Proof. Let $f_0 : 0 \to M$ and $g_0 : M'' \to 0$.

(1) Exactness at M' means $\operatorname{im} f_0 = \ker f$. Since $\operatorname{im} f_0 = \{0\}$, this is equivalent to $\ker f = \{0\}$, which is equivalent to f being injective.

(2) Exactness at M'' means $\operatorname{im} g = \ker g_0$. Since g_0 is the zero map to the zero module, we have $\ker g_0 = M''$. Thus exactness is equivalent to $\operatorname{im} g = M''$, which is equivalent to g being surjective.

□

From the proposition above, we say a sequence

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0. \tag{3}$$

is exact if and if only f is injective, g is surjective. Moreover, g induces an isomorphism[6] of $\operatorname{coker} f = M/f(M')$ onto M''.

A sequence of type (3) is called a **short exact sequence, S.E.S**[7].

---

[1] 列 [2] A-加群 [3] A-準同型写像 [4] M_i で完全 [5] 基本 [6] This is obvious because $\operatorname{coker} f = M/\operatorname{im} f = M/\ker g \cong \operatorname{im} g = M''$. [7] 短完全列

Proposition 2.2

Any long exact sequence of type (0) can be split up into short exact sequences.
If $N_i = \mathrm{im}(f_i) = \ker(f_{i+1})$, we have short exact sequences

$$0 \longrightarrow N_i \longrightarrow M_i \longrightarrow N_{i+1} \longrightarrow 0 \tag{3'}$$

for each i.

We will prove that (3') is indeed an exact sequence.

Proof. First, consider the inclusion $\mathfrak{i} : N_i \hookrightarrow M_i$. This inclusion is injective.
On the other hand, by the isomoprhism theorem for f_{i+1}:

$$\overline{f_{i+1}} : M_i/N_i \cong N_{i+1}.$$

Then, the map

$$F = \overline{f_{i+1}} \circ \pi_i : M_i \to N_{i+1}$$

is surjective (here, $\pi_i : M_i \to M_i/N_i$ is the natural projection of M_i). □

Proposition 2.3 (2.9)

i) Let

$$M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0 \tag{4}$$

be a sequence of A-modules and homomorphisms. Then the sequence (4) is exact if and only if for all A-modules N the sequence

$$0 \longrightarrow \mathrm{Hom}(M'', N) \xrightarrow{v_*} \mathrm{Hom}(M, N) \xrightarrow{u_*} \mathrm{Hom}(M', N) \tag{4'}$$

is exact.

ii) Let

$$0 \longrightarrow N' \xrightarrow{u} N \xrightarrow{v} N'' \tag{5}$$

be a sequence of A-modules and homomorphisms. Then the sequence (5) is exact if and only if for all A-modules M the sequence

$$0 \longrightarrow \mathrm{Hom}(M, N') \xrightarrow{u_*} \mathrm{Hom}(M, N) \xrightarrow{v_*} \mathrm{Hom}(M, N'') \tag{5'}$$

is exact.

Remark (For the Category Theory's Maniac) — The reversal of arrows from Proposition 2.3 follows directly from the functorial nature of Hom.

In the first argument, Hom(M, –) is a **covariant functor**. The morphism

$$v : N' \to N,$$

induces

$$v_* : \mathrm{Hom}(M, N') \to \mathrm{Hom}(M, N), \quad f \mapsto v \circ f.$$

Thus the direction is preserved, as seen in passing from (5) to (5').

In contrast, Hom(–, N) in the second argument is a **contravariant functor**. The morphism

$$u : M' \to M$$

induces

$$u_* : \mathrm{Hom}(M, N) \to \mathrm{Hom}(M', N), \quad f \mapsto f \circ u.$$

Thus, the direction of the arrow is reversed. This explains why sequence (4) leaves to (4') with arrow going in the opposite direction!

Alternatively, refer to the original book p.18 (or p.28 on the Japanese version)!

Proof. Brace yourself! the proof will be obnoxiously[8] long.
We will divide the proof into four parts: i) $\Rightarrow$); i) $\Leftarrow$); ii) $\Rightarrow$); and ii) $\Leftarrow$).

i) $\Rightarrow$) Suppose the sequence (4) is exact. For each A-modules N, we will prove that (4') is exact. That is, showing that a) $\ker u_* \supseteq \mathrm{im}\, v_*$; b) $\ker u_* \subseteq \mathrm{im}\, v_*$; c) v_* is injective:

a) Suppose $\phi \in \mathrm{im}\, v_*$. Our goal is to show that $u_*(\phi) = 0$.
There exists $\psi \in \mathrm{Hom}(M'', N)$ such that $\phi = v_*(\psi)$. Then,

$$u_*(\phi) = \phi \circ u = v_*(\psi) \circ u = (\psi \circ v) \circ u = \psi \circ (v \circ u) = 0.$$

The last equality comes from the fact that sequence (4) is exact.

b) Suppose $\phi \in \ker u_*$. Our goal is to show that $\phi \in \mathrm{im}\, v_*$.
Since v is surjective, for each $y \in M''$, there exists $x \in M$ such that $y = v(x)$. Define

$$\mathrm{Hom}(M'', N) \ni \psi : M'' \to N, \ \psi(y) := \phi(x).$$

Therefore, $\phi = \psi \circ v = v_*(\psi)$, which obviously shows that $\phi \in \mathrm{im}\, v_*$.

> **Remark** — ψ is well defined.
> If $v(x_1) = v(x_2)$, we have $x_1 - x_2 \in \ker v$. On the other hand, (4) is exact so $x_1 - x_2 \in \mathrm{im}\, u$. This means there exiss $a \in M'$ such that $u(a) = x_1 - x_2$. Therefore,
> $$\phi(x_1 - x_2) = \phi(u(a)) = 0$$
> yields $\phi(x_1) = \phi(x_2)$.

c) Suppose $v_*(\phi) = v_*(\psi)$ for some $\phi, \psi \in \mathrm{Hom}(M'', N)$. Our goal is to show that $\phi = \psi$.
Take any $y \in M''$. Since (4) is exact, v is surjective. Therefore, there exists $x \in M$ such that $v(x) = y$. Then

$$\phi(y) = \phi(v(x)) = v_*(\phi)(x) = v_*(\psi(x)) = \psi(y).$$

[8] 不快的

Hence, since y is arbitrary, $\psi = \phi$. We conclude that v_* is injective.

i) $\Leftarrow$) Assume for each $A-$modules N the sequence $(4')$ is exact.
We shall prove a) $\operatorname{im} u \subseteq \ker v$; b) $\operatorname{im} u \supseteq \ker v$; and c) v is surjective:

a) Our goal is to show that $v \circ u = 0$. That is, $v \circ u(x) = 0$ for every $x \in M'$.
Since $(4')$ is exact, $u_* \circ v_* = 0$. That is, for each $f : M'' \to N$, $f \circ v \circ u = 0$. Since N is arbitrary, we can choose N to be M'' and f to be $\mathrm{id}_{M''}$ so that

$$0 = \mathrm{id}_{M''} \circ v \circ u\circ = v \circ u.$$

b) Next, Our goal is to show that $x \in \operatorname{im} u$ whenever $x \in \ker v$.
Preparation. Since N is arbitrary, we can choose N to be $M/\operatorname{im} u$ and define $\phi : M \to N$ to be the natural projection of M. Then, we have

$$\phi \circ u = 0.$$

This means, $u_*(\phi) = 0$. Thus $\phi \in \ker u_*$. On the other hand, since $(4')$ is exact, $\phi \in \operatorname{im} v_*$. This means, there exists

$$\psi : M'' \to N \text{ satisfying } v_*(\psi) = \phi.$$

End of preparation.
Now take $x \in \ker v$. We have

$$\phi(x) = v_*(\psi)(x) = \psi \circ v(x) = \phi(0) = 0.$$

This implies $x \in \operatorname{im} u$ since ϕ is a map from M to $M/\operatorname{im} u$.

c) Since (4') is exact, v_* is injective.
Claim. If v_* is injective, then v is surjective.
Proof of the claim. Assume v is **not** surjective. Then, $\operatorname{im} v \subsetneq M''$. We can define $N := M''/\operatorname{im} v$ and a natural projection

$$\psi : M'' \to N.$$

If we can show that $[\psi \neq 0$ and $v_*(\psi) = 0]$, then v_* is **not** injective:

- $\operatorname{im} v \neq M''$ thus $\psi \neq 0$.
- For each $x \in M$,

$$v_*(\psi)(x) = \psi \circ v(x) = \psi(v(x)) = \psi(0) = 0,$$

since $v(x) \in \operatorname{im} v$.

ii) $\Rightarrow$) Suppose the sequence (5) is exact. For each A-modules M, we will prove that (5') is exact.
That is, showing that a) $\ker v_* \supseteq \operatorname{im} u_*$; b) $\ker v_* \subseteq \operatorname{im} u_*$; c) u_* is surjective:

a) Let $\phi \in \operatorname{im} u_*$. Then there exists $\psi \in \operatorname{Hom}(M, N')$ such that $\phi = u_*(\psi) = u \circ \psi$. Hence

$$v_*(\phi) = v \circ \phi = v \circ u \circ \psi = 0,$$

because $v \circ u = 0$. Thus $\phi \in \ker v_*$.

b) Let $\phi \in \ker v_*$. Then

$$v \circ \phi = 0.$$

Thus for every $x \in M$, we have $\phi(x) \in \ker v = \operatorname{im} u$. Since $u : N' \to N$ is injective, for each $x \in M$ there exists a unique element $\psi(x) \in N'$ such that

$$u(\psi(x)) = \phi(x).$$

This defines a homomorphism $\psi : M \to N'$. Therefore

$$u_*(\psi) = u \circ \psi = \phi,$$

so $\phi \in \operatorname{im} u_*$. Hence (5') is exact.

c) Suppose $u_*(\phi) = u_*(\psi)$ for some $\phi, \psi \in \operatorname{Hom}(M, N')$. Then

$$u \circ \phi = u \circ \psi.$$

Since u is injective, it follows that $\phi = \psi$. Hence u_* is injective.

ii) $\Leftarrow$) Assume for each $A-$modules M the sequence (5′) is exact.
We shall prove a) $\operatorname{im} u \subseteq \ker v$; b) $\operatorname{im} u \supseteq \ker v$; and c) v is surjective:

a) Our goal is to show that $v \circ u = 0$.
Since (5') is exact, $v_* \circ u_* = 0$. That is, for every $f : M \to N'$, $v \circ u \circ f = 0$. Since M is arbitrary, we can choose M to be N' and f to be $\mathrm{id}_{N'}$ so that $0 = v \circ u \circ \mathrm{id}_{N'} = v \circ u$.

b) Let $x \in \ker v$. We want to show that $x \in \operatorname{im} u$. Choose $M = A$ and define $\phi : A \to N$ by

$$\phi(a) = ax.$$

Since $x \in \ker v$, we have $v \circ \phi = 0$. Thus $\phi \in \ker v_*$. Since (5') is exact, $\ker v_* = \operatorname{im} u_*$. Hence there exists $\psi : A \to N'$ such that $u_*(\psi) = \phi$. Therefore

$$u(\psi(1)) = \phi(1) = x.$$

Hence $x \in \operatorname{im} u$.

c) Since (5') is exact, u_* is injective for every M. Choose $M = \ker u$ and let $i : \ker u \to N'$ be the inclusion map. Then

$$u_*(i) = u \circ i = 0.$$

Since u_* is injective, we get $i = 0$. Hence $\ker u = 0$. Therefore u is injective.

□

Proposition 2.4 (2.10)

Let

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f''} & & \\ 0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' & \longrightarrow & 0 \end{array}$$

be a commutative diagram of A-modules and homomorphisms, with the rows exact. Then there exists an exact sequence

$$\begin{aligned} 0 \longrightarrow \ker f' \xrightarrow{u_*} \ker f \xrightarrow{v_*} \ker f'' \xrightarrow{\partial} \\ \operatorname{coker} f' \xrightarrow{u'_*} \operatorname{coker} f \xrightarrow{v'_*} \operatorname{coker} f'' \longrightarrow 0, \end{aligned} \tag{6}$$

in which u_*, v_* are restrictions[a] of u, v, and u'_*, v'_* are induced[b] by u', v' (respectively).[c]

[a] 制限
[b] 誘導
[c] For example, $u_* := u|_{\ker f'}$

Proof. The boundary homomorphism[9] ∂ is defined as follows. Choose $x'' \in \ker f''$. Since v is surjective, there exists $x \in M$ such that $v(x) = x''$. Next, since the diagram is commutative,

$$v' \circ f(x) = f'' \circ v(x) = 0.$$

This implies $f(x) \in \ker v'$. Since the row is exact, $f(x) \in \operatorname{im} u'$. Consequently, there exists and element $x' \in N'$ such that $f(x) = u'(x')$. This motivates us to define

$$\partial(x'') := [x'] \in \operatorname{coker} f' = N' / \operatorname{im} f'.$$

Claim. ∂ is well defined.

Remark — So far we define ∂ as follows:

$$\text{For each } x'' \in \ker f'' \rightsquigarrow \underbrace{\textbf{choose } x \in M}_{:=\text{ b)}} \rightsquigarrow \underbrace{\textbf{choose } x' \in N'}_{:=\text{ a)}} \rightsquigarrow [x'].$$

Therefore, to prove the claim, we need to show that ∂ is independent of the choices of x and x'[a]!

[a] ∂ は x と x' の代表の取り方によらないことを示す.

Proof of the claim. Repeating the same argument above, we have

$$\begin{aligned} v(x) = x'', f(x) = u'(x'), \quad x \in M, x' \in N', \\ v(\tilde{x}) = x'', f(\tilde{x}) = u'(\tilde{x}'), \quad \tilde{x} \in M, \tilde{x}' \in N'. \end{aligned}$$

a) This case is easy. For $x \in M$ (the same argument works for $\tilde{x} \in M$), suppose there are x' and y' such that $f(x) = u'(x') = u'(y')$. Since u' is injective, we have $x' = y'$.

b) Now, we are going to show that for every $x, \tilde{x} \in M$, $v(x) = v(\tilde{x})$ we have $[x'] = [\tilde{x}']$. It is enough to show that $x' - \tilde{x}' \in \operatorname{im} f'$.

[9] 境界準同型写像

Since $v(x) = v(\tilde{x})$, we have $x - \tilde{x} \in \ker v = \operatorname{im} u$. Then, there exists $m' \in M'$ such that $x - \tilde{x} = u(m')$. Therefore,

$$u'(x - \tilde{x}') = f(x - \tilde{x}) = f \circ u(m') = u' \circ f'(m'),$$

where the final equality comes from the fact that the diagram is commutative. Therefore, since u' is injective, we have

$$x - \tilde{x}' \in \operatorname{im} f'.$$

End of proof of the claim

Now, it remains to prove the exactness of sequence (6).

$\ker \partial = \operatorname{im} v_*$) First, take $x'' \in \ker \partial$. We will show that $x'' \in \operatorname{im} v_*$.
Since v is surjective, there exists $x \in M$ such that $x'' = v(x)$. Then,

$$v' \circ f(x) = f'' \circ v(x) = 0,$$

where the first equality comes from the commutative diagram. We have $f(x) \in \ker v' = \operatorname{im} u'$. Consequently, there exists $y' \in N'$ such that $f(x) = u'(y')$. Therefore, by the definition of ∂ we have

$$0 = \partial(x'') = [y'].$$

This means $y' \in \operatorname{im} f'$, so there exists $z' \in M'$ such that $y' = f'(z')$. Now, consider

$$f(x - u(z')) = f(x) - f \circ u(z') = f(x) - u' \circ f'(z') = f(x) - f(x) = 0.$$

This implies $x - u(z') \in \ker f$. This allow us to map $x - u(z')$ through v_*. Finally,

$$\operatorname{im} v_* \ni v_*(x - u(z')) = v(x - u(z')) = v(x) - v \circ u(z') = x'' - 0.$$

Now, we prove the other direction. Take $b \in \operatorname{im} v_*$. Our goal is to show that $b \in \ker \partial$.
There exists $a \in \ker f$ such that $b = v_*(a) = v(a)$. Then, $f(a) = 0 = u'(0)$ so by the definition of ∂, we have $\partial(b) = [0]$. This shows that $b \in \ker \partial$.

$\ker v_* = \operatorname{im} u_*$) First, take $a \in \ker v_*$. We are going to show that $a \in \operatorname{im} u_*$.
Notice that $a \in \ker v = \operatorname{im} u$ so there exists $m' \in M'$ such that $a = u(m')$. Next,

$$u' \circ f'(m') = f \circ u(m') = 0.$$

Therefore, $f'(m') \in \ker u'$. However, u' is injective so $f'(m') = 0$. Therefore, $m' \in \ker f'$. Hence, we have shown that

$$\operatorname{im} u_* \ni a = u(m') = u_*(\underbrace{m'}_{\in \ker f'}).$$

Now, we prove the other direction. Take $a \in \operatorname{im} u_*$. There exists $b \in \ker f'$ such that $a = u_*(b) = u(b)$. Then,

$$v_*(a) = v(a) = v \circ u(b) = 0,$$

where the first equation comes from the fact that $a \in \ker f$ This shows that $a \in \ker v_*$.

u_* injective) Obvious from the fact that u is injective.

$\ker u'_* = \operatorname{im}\partial$) Take $[a] \in \ker u'_* \subseteq \operatorname{coker} f'$. This means that $a \in N'$, $u'_*([a]) = [u'(a)] = 0$ in the $\operatorname{coker} f$. our goal is to show that $[a] \in \operatorname{im}\partial$.
From $[u'(a)] = 0$ we have $u'(a) \in \operatorname{im} f$. Then, there exists $m \in M$ such that $f(m) = u'(a)$. Now, $v(m) \in \ker f''$ because

$$f'' \circ v(m) = v' \circ f(m) = v' \circ u'(a) = 0,$$

where the first equation comes from commutative diagram and the final equation comes from the exactness S.E.S. Therefore, by the definition of ∂ we can conclude

$$[a] = \partial(v(m)) \in \operatorname{im}\partial.$$

Now we prove the other direction. Take $x'' \in \ker f''$ and $\partial(x'') = [a] \in \operatorname{im}\partial$. Our goal is to prove $[a] \in \ker u'_*$.
From the definition of ∂, there exists $m \in M$ such that

$$v(m) = x'', \ f'(m) = u'(a).$$

Now,

$$u'_*([a]) = [u'(a)] = [f(m)] = 0 \text{ in the } \operatorname{coker} f,$$

where the last equation comes from the fact that $f(m) \in \operatorname{im} f$. Therefore, $[a] \in \ker u'_*$.

v'_* surjective) Consider the map

$$\begin{aligned} v'_* : \ \operatorname{coker} f &\longrightarrow \operatorname{coker} f'' \\ [x] &\longmapsto [v(x)]. \end{aligned}$$

The map is well defined (see remark below!). Take an arbitrary $[z''] \in \operatorname{coker} f''$ with $z'' \in N''$. Since v' is surjective, there exists $y \in N$ such that $v'(y) = z''$. Therefore,

$$v'_*([y]) = [v'(y)] = [z''].$$

□

Remark (The map between coker たち) — We will define the map of $\operatorname{coker} f \to \operatorname{coker} f''$. From the diagram, we have $v : M \to M''$ and $v' : N \to N''$ with the property

$$f'' \circ v = v' \circ f.$$

We define

$$\begin{aligned} v'_* : \ \operatorname{coker} f &\longrightarrow \operatorname{coker} f'' \\ [x] &\longmapsto [v'(x)]. \end{aligned}$$

Exercise 2.5. Show that v'_* is well defined!

Let C be a class of A-modules and let λ be a function on C with values in $\mathbb{Z}$ (or, more generally, with values in an abelian group G). The function λ is said to be **additive**[10] if, for each short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{3}$$

in which all the terms belong to C, we have

$$\lambda(M') - \lambda(M) + \lambda(M'') = 0.$$

[10] 加法的

Example (?????)

Let A be a field k, and let C be the class of all finite-dimensional k-vector spaces V. Then

$$V \longmapsto \dim V$$

is an additive function on C.[a]

[a] The original writer of the book did a **horrible** job explaining why this in an additive function on C.

Consider a short exact sequence

$$0 \longrightarrow V' \xrightarrow{f} V \xrightarrow{g} V'' \longrightarrow 0,$$

where $V', V, V'' \in C$. The dimension function

$$\dim : C \to \mathbb{Z}$$

is additive. In particular,

$$0 = \dim V' - \dim V + \dim V''.$$

Proof. Exactness gives $\ker g = \operatorname{im} f \cong V'$ (since f is injective) and $\operatorname{im} g = V''$. By the First Isomorphism Theorem,

$$V/\ker g \cong \operatorname{im} g = V'',$$

hence $V/V' \cong V''$. Therefore,

$$\dim V = \dim V' + \dim(V/V') = \dim V' + \dim V'',$$

which proves the claim. □

Proposition 2.6

Let

$$0 \longrightarrow M_0 \longrightarrow M_1 \longrightarrow \cdots \longrightarrow M_n \longrightarrow 0$$

be an exact sequence of A-modules in which all the modules M_i and the kernels of all the homomorphisms belong to C. Then for any additive function λ on C we have

$$\sum_{i=0}^{n} (-1)^i \lambda(M_i) = 0.$$

Proof. By proposition 2.2, we can split up the sequence into short exact sequences

$$0 \longrightarrow N_i \longrightarrow M_i \longrightarrow N_{i+1} \longrightarrow 0$$

with $N_i = \ker f_{i+1}$, $N_0 = N_{n+1} = 0$. Then we have

$$\lambda(M_i) = \lambda(N_i) + \lambda(N_{i+1}).$$

Therefore,

$$\sum_{i=0}^{n} (-1)^i \lambda(M_i) = \begin{matrix} \lambda(N_0) & -\lambda(N_1) & +\lambda(N_2) & -\lambda(N_3) & \cdots & +(-1)^n \lambda(N_n) \\ \lambda(N_1) & -\lambda(N_2) & +\lambda(N_3) & -\lambda(N_4) & \cdots & +(-1)^n \lambda(N_{n+1}) \end{matrix} = 0.$$

□

Example

The alternating sum gives a quick way to check computations in the Mayer–Vietoris long exact sequence. For example, suppose that after computing a Mayer–Vietoris sequence, we obtain

$$\begin{aligned} 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \longrightarrow \\ \mathbb{Z} \oplus \mathbb{Z} \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow 0. \end{aligned}$$

Then we can tell that this sequence is **wrong**, because by Proposition 2.5,

$$0 - 1 + 2 - 1 + 2 - 2 + 3 - 1 = 2 \neq 0.$$

3

July, 22 2026

Example 1) Let $\mathfrak{p} \subseteq A$ be a prime ideal of A. Then, for $S = A - \mathfrak{p}$ we have

$$S^{-1}A := A_{\mathfrak{p}}.$$

Moreover, $A_{\mathfrak{p}}$ is a local ring.

The process of passing from A to $A_{\mathfrak{p}}$ is called localization at $\mathfrak{p}$[a].

2) $S^{-1}A = 0 \iff 0 \in S$

3) Let $f \in A$ and $S := \{f^n\}_{n \geq 0}$. We write $S^{-1}A := A_f$.

4) Let $\mathfrak{a} \subseteq A$ be ideal in A and let

$$S = 1 + \mathfrak{a} = \{1 + x \mid x \in \mathfrak{a}\}.$$

Then S is multiplicatively closed.

[a] A から $A_{\mathfrak{p}}$ へ移行する操作を $\mathfrak{p}$ における局所化 (localization) という.

1) First we notice[1] that

$$S = A - \mathfrak{p} \text{ is multiplicatively closed set} \iff \mathfrak{p} \text{ is prime ideal in A.}$$

One can verify[2] that the assertion[3] is true.
Now, define

$$\mathfrak{m} := \{a/s \in A_{\mathfrak{p}} \mid a \in \mathfrak{p}, s \notin \mathfrak{p}\}.$$

Claim. $\mathfrak{m}$ is the only maximal ideal[4] of $A_{\mathfrak{p}}$.

Proof. First, recall proposition 1.6:

Proposition 3.1 (1.6)
Let A be a ring and $\mathfrak{m} \neq (1)$ an ideal of A such that every $x \in A - \mathfrak{m}$ is a unit in A. Then A is a local ring and $\mathfrak{m}$ its maximal ideal.

- ($\mathfrak{m}$ is an ideal of $A_{\mathfrak{p}}$)
  For $a/s, a'/s' \in \mathfrak{m}$ (notice that $a, a' \in \mathfrak{p}, s, s' \notin \mathfrak{p}$),

  $$a/s + a'/s' = (as' + a's)/ss' \in \mathfrak{m}$$

  because $as' + a's \in \mathfrak{p}$ and $ss' \notin \mathfrak{p}$.

[1] 気づく [2] 確認 [3] 主張 [4] $\mathfrak{m}$ is often written as $\mathfrak{p}A_{\mathfrak{p}}$

- (Every $x \in A_{\mathfrak{p}} - \mathfrak{m}$ is a unit in $A_{\mathfrak{p}}$)
  For each $x = b/t \in A_{\mathfrak{p}} - \mathfrak{m}$, since $b \notin \mathfrak{p}$, we have $b \in S$ and thus b/t is a unit in $A_{\mathfrak{p}}$.

□

2) If $S^{-1}A = 0$ (e.g it is a zero ring), then we see that

$$1/1 = 0/1.$$

So there exists $t \in S$ such that

$$t(1 \cdot 1 - 1 \cdot 0) = 0.$$

This implies $0 = t \in S$.

Conversely, if $0 \in S$, then for each $a/s \; in S^{-1}A$ we choose $t = 0 \in S$ so that

$$t(a \cdot 1 - 0 \cdot s) = 0.$$

This implies $a/s = 0/1$. Since a/s is arbitrary[5], then S^{-1A} is the zero ring.

So far we defined ring of fraction $S^{-1}A$ from a ring A.

We can apply the same thing from a module M to obtain a **module of fraction** $S^{-1}M$. Define a relation $\sim$ on $M \times S$ as follows:

$$(m, s) \sim (m', s') \iff \exists t \in S \text{ such that } t(sm' - s'm) = 0.$$

This is indeed an equivalence relation. Denote the equivalence class of the pair (m, s) to be m/s. The set of such fraction is $S^{-1}M$. As examples 1) and 3) above, we write $S^{-1}M = M_p$ when $S = A - \mathfrak{p}$[6] prime and M_f when $S = \{f^n\}_{n \geq 0}$.

Let $u : M \to N$ be A-module homomorphism. Then it gives rise[7] to an $S^{-1}A$-module homomorphism

$$S^{-1}u : S^{-1}M \to S^{-1}N, \quad m/s \mapsto u(m)/s.$$

Exercise 3.2. Show that $S^{-1}u$ is well defined!

Furthermore, we have

$$S^{-1}(v \circ u) = (S^{-1}v) \circ (S^{-1}u).$$

Proposition 3.3 (3.3)

The operation S^{-1} is exact. That is, if

$$M' \xrightarrow{f} M \xrightarrow{g} M''$$

is exact at M, then

$$S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M''$$

is exact at $S^{-1}M$.

[5] 任意　[6] $S = M - \mathfrak{p}$ でないことに注意しよう！　[7] 引き起こす

Proof. We have $g \circ f = 0$, hence $S^{-1}g \circ S^{-1}f = S^{-1}(0) = 0$, hence $\mathrm{Im}(S^{-1}f) \subset \ker(S^{-1}g)$.
To prove the reverse inclusion, let $m/s \in \ker(S^{-1}g)$; then

$$g(m)/s = 0 \text{ in } S^{-1}M'',$$

hence there exists $t \in S$ such that $tg(m) = 0$ in M''. But $tg(m) = g(tm)$ since g is an A-module homomorphism, hence $tm \in \ker g = \mathrm{im}\, f$, and therefore $tm = f(m')$ for some $m' \in M'$. Hence in $S^{-1}M$ we have

$$m/s = f(m')/st = (S^{-1}f)(m'/st) \in \mathrm{im}(S^{-1}f).$$

□

In particular, it follows from proposition 3.3 that if M' is a submodule of M, the mapping $S^{-1}M' \to S^{-1}M$ is injective and therefore $S^{-1}M'$ can be regarded as a submodule of $S^{-1}M$[8]. With these relation in mind, we obtain the following result:

Corollary 3.4 (3.4)

Formation of fractions commutes with formation of finite sums, finite intersections and quotients.

Precisely, if N, P are submodules of an A-module M, then

i) $S^{-1}(N + P) = S^{-1}(N) + S^{-1}(P)$;

ii) $S^{-1}(N \cap P) = S^{-1}(N) \cap S^{-1}(P)$;

iii) the $S^{-1}A$-modules $S^{-1}(M/N)$ and $(S^{-1}M)/(S^{-1}N)$ are isomorphic.

Proof.

i) We show that $S^{-1}(N + P) \supseteq S^{-1}(N) + S^{-1}(P)$.
Take $n/s \in S^{-1}N$ and $p/s' \in S^{-1}P$. Notice that

$$S^{-1}N + S^{-1}P \ni n/s + p/s' = (ns' + ps)/ss' \in S^{-1}(N + P).$$

The reverse direction is obvious.

ii) We show that $S^{-1}N \cap S^{-1}P \subset S^{-1}(N \cap P)$.
If $y/s = z/t$ with $y \in N$, $z \in P$, $s, t \in S$, then

$$u(ty - sz) = 0 \text{ for some } u \in S,$$

hence $w := tuy = usz \in N \cap P$ and therefore $y/s = w/stu \in S^{-1}(N \cap P)$. Consequently

$$S^{-1}N \cap S^{-1}P \subset S^{-1}(N \cap P).$$

The reverse direction is obvious.

iii) Apply S^{-1} to the exact sequence

$$0 \to N \to M \to M/N \to 0.$$

We obtain

$$0 \to S^{-1}N \to S^{-1}M \to S^{-1}(M/N) \to 0.$$

A little bit of diagram chase we obtain the assertion.

□

[8] M' が M の部分加群ならば，写像 $S^{-1}M' \to S^{-1}M$ は単射であるから，$S^{-1}M'$ は $S^{-1}M$ の部分加群とみることができる

Proposition 3.5

Let M be an A-module. Then the $S^{-1}A$-modules $S^{-1}A \otimes_A M$ and $S^{-1}M$ are isomorphic; more precisely, there exists a unique isomorphism

$$f' : S^{-1}A \otimes_A M \to S^{-1}M \tag{1}$$

for which $f'\big((a/s) \otimes m\big) = am/s$ for all $a \in A$, $m \in M$, $s \in S$.

Proof. Define $f : S^{-1}A \times M \to S^{-1}M$ by

$$(a/s, m) \mapsto am/s.$$

f is A-bilinear, and therefore by the universal property (proposition 2.12) of the tensor product induces an A-homomorphism

$$f' : S^{-1}A \otimes_A M \to S^{-1}M$$

satisfying (1). Clearly f' is surjective, and is uniquely defined by (1).

Claim. Hence every element of $S^{-1}A \otimes_A M$ is of the form $(1/s) \otimes m$. (see remarks after the proof)

Assume the claim is true. We show that f' is injective.

Suppose that

$$f\big((1/s) \otimes m\big) = 0.$$

Then $m/s = 0$, hence $tm = 0$ for some $t \in S$, and therefore

$$\frac{1}{s} \otimes m = \frac{1}{st} \otimes tm = \frac{1}{st} \otimes 0 = 0.$$

Hence f' is injective and therefore an isomorphism. □

Remark — Let $\sum_i (a_i/s_i) \otimes m_i$ be any element of $S^{-1}A \otimes_A M$. If $s = \prod_i s_i \in S$, $t_i = \prod_{j \neq i} s_j$, we have

$$\sum_i \frac{a_i}{s_i} \otimes m_i = \sum_i \frac{a_i t_i}{s} \otimes m_i = \sum_i \frac{1}{s} \otimes a_i t_i m_i = \frac{1}{s} \otimes \sum_i a_i t_i m_i.$$

Therefore, every element of $S^{-1}A \otimes_A M$ is of the form $(1/s) \otimes m$!

Corollary 3.6

$S^{-1}A$ is a flat A-module.

Proof. (Sketch only)

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

By proposition 3.3:

$$0 \longrightarrow S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M'' \longrightarrow 0$$

By proposition 3.5:

$$0 \longrightarrow S^{-1}A \otimes_A M' \longrightarrow S^{-1}A \otimes_A M \longrightarrow S^{-1}A \otimes_A M'' \longrightarrow 0.$$

□

Proposition 3.7

If M, N are A-modules, there is a unique isomorphism of $S^{-1}A$-modules

$$f : S^{-1}M \otimes_{S^{-1}A} S^{-1}N \to S^{-1}(M \otimes_A N)$$

such that

$$f\big((m/s) \otimes (n/t)\big) = (m \otimes n)/st.$$

In particular, if $\mathfrak{p}$ is any prime ideal, then

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes_A N)_{\mathfrak{p}}$$

as $A_{\mathfrak{p}}$-modules.

Proof. Notice that

$$\begin{aligned}
S^{-1}M \otimes_{S^{-1}A} S^{-1}N &\cong (S^{-1}A \otimes_A M) \otimes_{S^{-1}A} (S^{-1}A \otimes_A N) && \text{by Proposition 3.5}\\
&\cong (M \otimes_A S^{-1}A) \otimes_{S^{-1}A} (S^{-1}A \otimes_A N) && \text{by Proposition 2.14(i)}\\
&\cong M \otimes_A (S^{-1}A \otimes_{S^{-1}A} (S^{-1}A \otimes_A N)) && \text{by Exercise 2.15}\\
&\cong M \otimes_A ((S^{-1}A \otimes_{S^{-1}A} S^{-1}A) \otimes_A N) && \text{by Exercise 2.15}\\
&\cong M \otimes_A (N \otimes_A (S^{-1}A \otimes_{S^{-1}A} S^{-1}A)) && \text{by Proposition 2.14(i)}\\
&\cong (M \otimes_A N) \otimes_A (S^{-1}A \otimes_{S^{-1}A} S^{-1}A) && \text{by Proposition 2.14(ii)}\\
&\cong (S^{-1}A \otimes_{S^{-1}A} S^{-1}A) \otimes_A (M \otimes_A N) && \text{by Proposition 2.14(i)}\\
&\cong S^{-1}A \otimes_A (M \otimes_A N) && \text{by Proposition 2.14(iv)}\\
&\cong S^{-1}(M \otimes_A N) && \text{by Proposition 3.5}
\end{aligned}$$

□

§3.1 Local Properties (3.2)

A property P of a ring A (or of an A-module M) is said to be a **local property** if the following is true:

$$A \text{ (or } M\text{) has } P \iff A_{\mathfrak{p}} \text{ (or } M_{\mathfrak{p}}\text{) has } P,$$

for each prime ideal $\mathfrak{p}$ of A.

Proposition 3.8

Let M be an A-module. Then the following are equivalent:

i) $M = 0$;

ii) $M_{\mathfrak{p}} = 0$ for all prime ideals $\mathfrak{p}$ of A;

iii) $M_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$ of A.

Proof. Clearly i) ⇒ ii) ⇒ iii). Suppose iii) satisfied and $M \neq 0$. Let x be a non-zero element of M, and let

$$\mathfrak{a} = \operatorname{Ann}(x) = \{a \in A \mid ax = 0\}.$$

$\mathfrak{a}$ is an ideal $\neq (1)$, hence is contained in a maximal ideal $\mathfrak{m}$ by corollary 1.4. Consider $x/1 \in M_\mathfrak{m}$. Since $M_\mathfrak{m} = 0$ we have $x/1 = 0$, hence x is killed by some element of $A - \mathfrak{m}$; but this is impossible since $\mathrm{Ann}(x) \subset \mathfrak{m}$. □

Proposition 3.9

Let $\varphi : M \to N$ be an A-module homomorphism. Then the following are equivalent:

i) φ is injective;

ii) $\varphi_\mathfrak{p} : M_\mathfrak{p} \to N_\mathfrak{p}$ is injective for each prime ideal $\mathfrak{p}$;

iii) $\varphi_\mathfrak{m} : M_\mathfrak{m} \to N_\mathfrak{m}$ is injective for each maximal ideal $\mathfrak{m}$.

Similarly with "injective" replaced by "surjective" throughout.

Proof.

i) ⇒ ii) $0 \to M \to N$ is exact, hence $0 \to M_\mathfrak{p} \to N_\mathfrak{p}$ is exact, i.e. $\varphi_\mathfrak{p}$ is injective.

ii) ⇒ iii) because a maximal ideal is prime.

iii) ⇒ i) Let $M' = \ker(\varphi)$, then the sequence $0 \to M' \to M \to N$ is exact, hence $0 \to M'_\mathfrak{m} \to M_\mathfrak{m} \to N_\mathfrak{m}$ is exact by proposition 3.3, and therefore

$$M'_\mathfrak{m} = \ker(\varphi_\mathfrak{m}) = 0$$

since $\varphi_\mathfrak{m}$ is injective. Hence $M' = 0$ by proposition 3.8, and therefore φ is injective.

For the other part of the proposition, just reverse all the arrows. □

Flatness is a local property:

Proposition 3.10

For any A-module M, the following statements are equivalent:

i) M is a flat A-module;

ii) $M_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-module for each prime ideal $\mathfrak{p}$;

iii) $M_\mathfrak{m}$ is a flat $A_\mathfrak{m}$-module for each maximal ideal $\mathfrak{m}$.

Proof.

i) ⇒ ii) See proposition 3.5 and exercise 2.20.

ii) ⇒ iii) Obvious.

iii) ⇒ i) If $N \to P$ is a homomorphism of A-modules, and $\mathfrak{m}$ is any maximal ideal of A, then

$$\begin{aligned} N \to P \text{ injective} &\Rightarrow N_\mathfrak{m} \to P_\mathfrak{m} \text{ injective, by proposition 3.9} \\ &\Rightarrow N_\mathfrak{m} \otimes_{A_\mathfrak{m}} M_\mathfrak{m} \to P_\mathfrak{m} \otimes_{A_\mathfrak{m}} M_\mathfrak{m} \text{ injective, by proposition 2.19} \\ &\Rightarrow (N \otimes_A M)_\mathfrak{m} \to (P \otimes_A M)_\mathfrak{m} \text{ injective, by proposition 3.7} \\ &\Rightarrow N \otimes_A M \to P \otimes_A M \text{ injective, by proposition 3.9} \end{aligned}$$

Hence M is flat by proposition 2.19.

□

4

November, 16 2026

§4.1 Valuation Rings (5.4)

Let us start from a definition.

Definition 4.1 (Valuation Ring)**.** Let B be an integral domain[1] and K its field of fraction[2]. We say that B is a **valuation ring**[3] of K if for each $x \neq 0$ (as the element of K![4]), either

$$x \in B \text{ or } x^{-1} \in B$$

or both.

Let us go through some examples:

Example

- Our old friend $B = \mathbb{Z}$ is **not** a valuation ring of $\mathbb{Q}$.
- Let p be a prime number. If

$$B = \mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; a, b \in \mathbb{Z},\ p \nmid b \right\},$$

then $K = \mathbb{Q}$. $\mathbb{Z}_{(p)}$ is a valuation ring of $\mathbb{Q}$.

Proposition 4.2 (5.18)

Let B be a valuation ring of K. Then, the followings are true:

i) B is a local ring[a].

ii) If B' is a ring such that $B \subseteq B' \subseteq K$, then B' is a valuation ring of K.

iii) B is integrally closed (in K)[b].

[a] 局所環
[b] B は K で整閉

Proof.

i) We recall a proposition:

[1] 整域
[2] 分数体
[3] 付値環
[4] 日本語版では、x が K の元であることが強調されていたが、英語版では特に強調されていなかった。なので、$x \in K$ だということに最初は気かなかった。

Proposition (Recall Proposition 1.6)

Let B be a ring, $\mathfrak{m} \neq (1)$ be an ideal of B such that for any $x \in B - \mathfrak{m}$, x is a unit[a] in B. Then, $\mathfrak{m}$ is the unique maximal ideal[b] of B and B is a local ring.

[a] 単元
[b] 極大イデアル

Now, similar to the 1.6, define a **set** $\mathfrak{m}$:

$$\mathfrak{m} := \{x \in B \mid x \text{ is not a unit in } B\}.$$

Then we have

$$x \in \mathfrak{m} \Leftrightarrow x = 0 \text{ or } x^{-1} \notin B.$$

We show that $\mathfrak{m}$ is an ideal of B:

1. For each $a \in B$ and $x \in \mathfrak{m}$, if $ax \notin \mathfrak{m}$ we have $(ax)^{-1} \in B$ so that
$$x^{-1} = a(ax)^{-1} \in B,$$
a contradiction[5].
2. For each $x, y \in \mathfrak{m}$, we show that $x + y \in \mathfrak{m} - 0$.
Consider $\frac{x}{y} \in K$. Then we have $xy^{-1} \in B$ or $x^{-1}y \in B$. If, for example $xy^{-1} \in B$, we have
$$x + y = (1 + xy^{-1})y \in B\mathfrak{m} \subseteq \mathfrak{m}.$$

Hence $\mathfrak{m}$ is an ideal (even better: maximal ideal) of B and therefore B is a local ring by Proposition 1.6.

ii) Clear from the definition.

iii) Let $x \in K$ be integral over[6] B. Then there exist $b_i \in B$ such that

$$x^n + b_1 x^{n-1} + \cdots + b_n = 0.$$

Now, if $x \in B$ we are done, so let $x^{-1} \in B$. Then

$$x = -(b_1 + b_2 x^{-1} + \cdots + b_n (x^{-1})^{n-1}) \in B.$$

□

Remark — From Proposition 4.2 (5.18) ii) we have

$$\text{valuation ring} \Longrightarrow \text{ local ring.}$$

The converse[a] is not true, in general.

[a] 逆が必ず成り立つとは限らない

[5] 矛盾
[6] 上で整である

Now, let K be a field, Ω an algebraically closed field[7]. Let

$$\Sigma = \{(A, f) \mid A \subseteq K \text{ subring}, f : A \to \Omega \text{ ring homomorphism}\}.$$

We particially order[8] the set Σ as follows:

$$(A, f) \leq (A', f') \iff A \subseteq A' \text{ and } f'|_A = f.$$

For each chain[9] $\sigma \subseteq \Sigma$, the element (A_σ, F) is an upper bound[10] of σ thus we can use Zorn's Lemma. Consequently[11], the set Σ has atleast one maximal element[12], say (B, g).

Our objective[13] is to show that B is a valuation ring of K. We need some preparation[14]:

Lemma 4.3 (5.19)

B is a local ring and $\mathfrak{m} = \ker g$ is its maximal ideal.

Proof. Notice that $g(B)$ is a subring of a field Ω (which is an integral domain). Thus $g(B)$ is also an integral domain. Therefore, the isomoprhism[15]

$$B/\ker g \cong g(B)$$

implies $\ker g$ a prime ideal[16]. Now, we can extend[17] g to a homomorphism

$$\overline{g} : B_{\mathfrak{m}} \to \Omega$$

via $\overline{g}(b/s) = g(b)/g(s)$ for all $b \in B$ and $s \in B - \mathfrak{m}$ (since $g(s)$ will not be zero). We obtain[18] $(B, g) \leq (B_{\mathfrak{m}}, \overline{g})$, and by the maximality[19] of (B, g) we obtain $B = B_{\mathfrak{m}}$. Hence B is a local ring and $\mathfrak{m}$ is its maximal ideal. □

Lemma 4.4 (5.20)

Let x be a non-zero element of K. Let $B[x]$ be the subring of K generated by x over B and let $\mathfrak{m}[x]$ be the extension[a] of $\mathfrak{m}$ in $B[x]$. Then, we have either $\mathfrak{m}[x] \neq B[x]$ or $\mathfrak{m}[x^{-1}] \neq B[x^{-1}]$.

[a] 拡大

Proof. Suppose the otherwise (i.e $\mathfrak{m}[x] = B[x]$ or $\mathfrak{m}[x^{-1}] = B[x^{-1}]$). Then since $1 \in B[x]$ and $1 \in B[x^{-1}]$, there exist $u_i, v_i \in \mathfrak{m}$ such that

$$u_0 + u_1 x + \cdots + u_m x^m = 1 \tag{1}$$
$$v_0 + v_1 x^{-1} + \cdots + v_n x^{-n} = 1, \tag{2}$$

[7] 代数的閉体 (つま, 一次以上の任意の K 係数一変数多項式が K 上に根を持つこと) [8] 半順序 [9] 鎖 (全順序部分集合) [10] 上界 [11] だから [12] 極大元 [13] 目標 [14] 準備 [15] 同型 [16] 素イデアル [17] 拡大する [18] 得る [19] 最大性

in which we assume m, n are as small as possible. WLOG $m \geq n$. Multiply (2) by x^n:

$$(1 - v_0)x^n = v_1 x^{n-1} + \cdots + v_n. \tag{3}$$

Now, since $v_0 \in \mathfrak{m}$, we claim that $1 - v_0 \notin \mathfrak{m}$. This is because if not, then $1 = (1 - v_0) + v_0 \in \mathfrak{m}$, a contradiction. Thus $1 - v_0$ is a unit in B. Therefore, (3) can be written in the form

$$x^n = w_1 x^{n-1} + \cdots + w_n,$$

where $w_i \in \mathfrak{m}$. Hence we can replace x^m in (1) by $w_1 x^{m-1} + \cdots + w_n x^{m-n}$, contradicting the minimality of the exponent[20] m. □

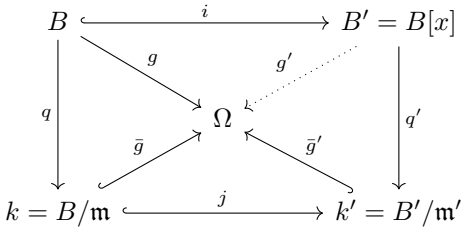
Now, here comes the main course[21]:

Theorem 4.5 (5.21)
Let (B, g) be a maximal element of Σ. Then B is a valuation ring of the field K.

Proof. Take a non-zero $x \in K$. Our goal it to prove $x \in B$.
Claim. There exists g' such that

$$(B, g) = (B', g').$$

Therefore, $x \in B[x] = B' = B$ implies $x \in B$.
Proof of the claim. (i.e. How do we construct g'?)
This is a diagram to help you understand the proof (it is not necessary commutative[22]!)



Step 1. From Lemma 4.4 (5.20) $\mathfrak{m}[x]$ is not the unit ideal[23] of ring $B' = B[x]$. Then, $\mathfrak{m}[x]$ is contained in a maximal ideal $\mathfrak{m}'$ of B' (i.e $\mathfrak{m}[x] \subseteq \mathfrak{m}' \subsetneq B'$).
Now we claim that $\mathfrak{m}' \cap B = \mathfrak{m}$. Indeed, since $\mathfrak{m}' \cap B$ is a proper ideal[24] of B[25], we have

$$\mathfrak{m} \subseteq \mathfrak{m}' \cap B \subsetneq B.$$

Since $\mathfrak{m}$ is the maximal ideal of B, then we have $\mathfrak{m} = \mathfrak{m}' \cap B$.

Now, write $k := B/\mathfrak{m}$ and $k' = B'/\mathfrak{m}'$. $\mathfrak{m}$ and $\mathfrak{m}'$ are maximal ideals so k and k' are fields.

The embedding[26] $i : B \hookrightarrow B'$ together with $\mathfrak{m} = \mathfrak{m}' \cap B$ induces[27] an embedding

$$j : k \hookrightarrow k' \text{ via } b + \mathfrak{m} \mapsto b + \mathfrak{m}'.$$

[20] 指数 [21] 主菜。ここでは、議論の核心を指す。[22] 可換 [23] 単位イデアル [24] 真のイデアル [25] Otherwise, we have $\mathfrak{m}' \cap B = B$, in particular $1 \in \mathfrak{m}'$, contradicting the maximality of $\mathfrak{m}'$. [26] 埋め込み [27] 誘導する

Indeed, j is injective[28]:

$$\ker j = \{b + \mathfrak{m} \mid b + \mathfrak{m}' \in \mathfrak{m}'\} = (B \cap \mathfrak{m}')/\mathfrak{m} = 0.$$

Step 2. Now, consider the natural map[29] $q' : B' \to k'$ where we identify the image[30] of x in k' as $\overline{x}$ (i.e. $\overline{x} := q'(x)$). We claim that $k' = k[\overline{x}]$.

Indeed, take $z \in k'$. Since q' is surjective[31], there exists $y \in B'$ such that $z = q'(y)$. We can write y as $y = a_0 + a_1 x + \cdots + a_n x^n$, where $a_i \in B$. Notice that $q'(a_i) = j(a_i + \mathfrak{m}) \in j(k) \cong k$:

$$\begin{aligned} z = q'(y) &= q'(a_0) + q'(a_1)q'(x) + \cdots + q'(a_n)q'(x)^n, \\ &= q'(a_0) + q'(a_1)\overline{x} + \cdots + q'(a_n)\overline{x}^n \in k[\overline{x}], \end{aligned}$$

hence $k' \subseteq k[\overline{x}]$.

Conversely, we know that $k \subseteq k'$ and $\overline{x} \in k'$ so $k[\overline{x}] \subseteq k'$. Moreover, x is algebraic over[32] k and therefore k' is a finite algebraic extension[33] of k.

Step 3. Now, the homomorphism $g : B \to k$ induces an embedding $\overline{g} : k \to \Omega$ (because $\mathfrak{m} = \ker g$). Furthermore, we extend $\overline{g} : k \to \Omega$ to an embedding $\overline{g}' : k' \to \Omega$.

How does one construct such embedding?

Since Ω is algebraically closed, every non-constant polynomial[34] in $\Omega(X)$ has root in Ω. Thus, when $\overline{x}$ is algebraic over k, the minimal polynomial[35] is induced via $\overline{g}$ to the polynomial $\Omega(X)$, then we can choose a root in Ω as the image of $\overline{x}$. Furthermore, since $k' = k[\overline{x}]$, every element in k' can be generated by k and an element $\overline{x}$.

Let

$$f(X) = a_0 + a_1 X + \cdots + a_n X^n \in k[X]$$

be the minimal polynomial of $\overline{x}$ over k (obviously $f(\overline{x}) = 0$). Now choose $\beta \in \Omega$ such that $\overline{g}'(\overline{x}) = \beta$. Since 0 goes to 0:

$$\begin{aligned} 0 = \overline{g}'(0) &= \overline{g}'(f(\overline{x})) \\ &= \overline{g}'(a_0) + \overline{g}'(a_1)\beta + \cdots + \overline{g}'(a_n)\beta^n. \end{aligned}$$

Therefore, β is a root of a polynomial

$$f^{\overline{g}'}(X) = \overline{g}'(a_0) + \overline{g}'(a_1)X + \cdots + \overline{g}'(a_n)X^n \in \Omega[X]$$

over Ω.

Finally, the embedding $\overline{g}'$ is defined as follows:

$$\overline{g}' : k' \to \Omega \text{ where } \overline{g}'|_k = g \text{ and } \overline{x} \mapsto \beta.$$

Step 4. Finally, we define $g' : B' \to \Omega$ via $g' = \overline{g}' \circ q'$ which extends g.

Since (B, g) is already the maximal element, we are left with $B = B'$. We are done. □

[28] 単射 [29] 自然な写像 [30] 像 [31] 全射 [32] 上で代数的である [33] 有限次代数拡大 [34] 定数でない多項式 [35] 最小多項式

Corollary 4.6 (5.22)

Let A be a subring of a field K. Then,

$$\overline{A} = \bigcap_{\substack{B \supseteq A \\ B\text{: valuation ring of } K}} B.$$

Here, $\overline{A}$ denotes the integral closure[a] of A in K.

[a] 整閉包

Proof. Let B be any valuation ring of K that contains A (i.e. $A \subseteq B$). Then $\overline{A} \subseteq \overline{B} = B$ by Proposition 4.2 (5.18) iii). Since B is arbitrary, we have

$$\overline{A} \subseteq \bigcap_{\substack{B \supseteq A \\ B\text{: valuation ring of } K}} B.$$

Conversely, If $x \notin \overline{A}$, it is enough to show that $x \notin B$. Notice that x is not in the ring $A' = A[x^{-1}]$. This is because if so, then there exist $a_i \in A$ such that

$$x = a_0 + a_1 x^{-1} + \cdots + a_n x^{-n}.$$

Thus x is integral over A (i.e. $x \in \overline{A}$), a contradiction. Hence x^{-1} is not a unit in A'. Therefore, x^{-1} is contained in the maximal ideal (say $\mathfrak{m}'$) of A'.

Now, let Ω be the algebraic closure[36] of the field $k' := A'/\mathfrak{m}'$ and define a homomoprhism $q' : A' \to k'$ and an inclusion[37] $i : k' \to \Omega$. We then consider a pair $(A', f) \in \Sigma$ with $f := i \circ q' : A' \to \Omega$. By Zorn's lemma, we have a maximal element $(B, g) \in \Sigma$ satisfying

$$(A', f) \leq (B, g).$$

This means $A' \subseteq B$ and $g|_{A'} = f$. By Theorem 4.5 (5.21), B is a valuation ring of K. Then, since $x^{-1} \in \mathfrak{m}' \subseteq A' \subseteq B$ and $q'(x^{-1}) = 0$ in $\mathfrak{m}'$ we have

$$g(x^{-1}) = g|_{A'}(x^{-1}) = i \circ q'(x^{-1}) = 0.$$

Finally, we have $x \notin B$. This is because if $x \in B$, $g(x)$ is defined and thus

$$1 = g(1) = g(x)g(x^{-1}) = g(x) \cdot 0 = 0,$$

a contradiction. □

Proposition 4.7 (5.23)

Let $A \subseteq B$ be integral domains, B finitely generated[a] over A. Let v be a non-zero element of B. Then there exists $u \neq 0$ in A with the following property: any homomorphism f of A into an algebraically closed field Ω such that $f(u) \neq 0$ can be extended to a homomorphism of B into Ω such that $g(v) \neq 0$.

[a] 有限生成

[36] 代数的閉包

[37] 包含写像

Proof. By induction[38] on the number of generators[39] of B over A we reduce immediately to the case where B is generated over A by a single element x.

i) Suppose x is transcendental[40] over A, i.e. that no non-zero polynomial with coefficients[41] in A has x as a root. Let $v = a_0x^n + a_1x^{n-1} + \cdots + a_n$, and take $u = a_0$. Then if $f\colon A \to \Omega$ is such that $f(u) \neq 0$, there exists $\alpha \in \Omega$ such that

$$f(a_0)\alpha^n + f(a_1)\alpha^{n-1} + \cdots + f(a_n) \neq 0,$$

because Ω is infinite. Define $g\colon B \to \Omega$ extending f by putting $g(x) = \alpha$. Then $g(v) \neq 0$, as required.

ii) Now suppose x is algebraic over A (i.e. over the field of fractions of A). Then, since v is a polynomial in terms of x, v is algebraic over A. In particular, v^{-1} is algebraic over A. Hence we have equations of the form

$$a_0x^m + a_1x^{m-1} + \cdots + a_m = 0 \qquad (a_i \in A), \tag{1}$$

$$a'_0v^{-n} + a'_1v^{-(n-1)} + \cdots + a'_n = 0 \qquad (a'_i \in A). \tag{2}$$

Let $u = a_0a'_0$ and let $f\colon A \to \Omega$ be such that $f(u) \neq 0$. Then f can be extended, first to a homomorphism $f' : A[u^{-1}] \to \Omega$ (with $f'(u^{-1}) = f(u)^{-1}$), and then by Theorem 4.5 (5.21) to a homomorphism $h\colon C \to \Omega$, where C is a valuation ring containing $A[u^{-1}]$.

From (1), we have

$$x^m + a_1a'_0u^{-1}x^{m-1} + \cdots + a_ma'_0u^{-1} = 0 \qquad (a_1a'_0u^{-1} \in A[u^{-1}])$$

hence x is integral over $A[u^{-1}]$, hence by Corollary 4.6 (5.22) $x \in C$, so that C contains B, and in particular $v \in C$. On the other hand, from (2), v^{-1} is integral over $A[u^{-1}]$, and therefore by Corollary 4.6 (5.22) again is in C. Therefore v is a unit in C, and hence $h(v) \neq 0$. Now take g to be the restriction[42] of h to B: $g(v) \neq 0$.

□

Corollary 4.8 (5.24)

Let k be a field and B a finitely generated k-algebra[a]. If B is a field then it is a finite algebraic extension of k.

[a] k-代数

Proof. Take $A = k$, $v = 1$ and Ω the algebraic closure of k and apply Proposition 4.7 (5.23). □

[38] 帰納法 [39] 生成元 [40] 超越的 [41] 係数 [42] 制限

5

未定 *unfinished*

§5.1 (Review) Primary Decomposition

A prime ideal in a ring A is motivated from the prime number. The corresponding generalization of a power of a prime number is a **primary ideal**.

Definition 5.1. Let A be a ring. an ideal $\mathfrak{q}$ in a ring A is **primary** if $\mathfrak{q} \neq A$ and if

$$xy \in \mathfrak{q} \Longrightarrow x \in \mathfrak{q} \text{ or } y^n \in \mathfrak{q} \text{ for some } n > 0.$$

In other words, we have the following proposition

Proposition 5.2

Let $\mathfrak{q}$ be an ideal in a ring A. Then, $\mathfrak{q}$ is primary if and if only $A/\mathfrak{q}$ is non-zero ring and every zero-divisor element in $A/\mathfrak{q}$ is nilpotent.

Proof. Let $\mathfrak{q}$ be primary. Then, since $\mathfrak{q} \neq A$, $A/\mathfrak{q}$ is not a zero ring. Take a zero-divisor $x+\mathfrak{q} \in A/\mathfrak{q}$ (of course, $x \notin \mathfrak{q}$). Then, there exists $y+\mathfrak{q} \in A/\mathfrak{q}$ ($y \notin \mathfrak{q}$) such that $xy \in \mathfrak{q}$. Since $\mathfrak{q}$ is primary, we have $x \in \mathfrak{q}$ or $y^n \in \mathfrak{q}$ for some positive integer n. Both $x \in \mathfrak{q}$ and $y^n \in \mathfrak{q}$ lead to a contradiction. Thus we have $y \in \mathfrak{q}$ or $x^n \in \mathfrak{q}$ for some positive integer n. Then, $(x+\mathfrak{q})^n = x^n + \mathfrak{q} \in \mathfrak{q}$. x is a nilpotent element.

Conversely, suppose $A/\mathfrak{q}$ is non-zero ring and every non zero-divisor element in $A/\mathfrak{q}$ is nilpotent. Choose an arbitrary element $x, y \in \mathfrak{q}$ such that $xy \in \mathfrak{q}$. If either x or y is zero, then result is trivial. So, suppose both x and y are non-zero. Then, x is a zero-divisor element in $A/\mathfrak{q}$ (and thus a nilpotent). Then, there exists positive integer n such that $x^n \in \mathfrak{q}$. □

Remark — Given an ideal $\mathfrak{q}$ of A, it is often useful to work in the quotient ring $A/\mathfrak{q}$ and use the correspondence between ideals of $A/\mathfrak{q}$ and ideals of A containing $\mathfrak{q}$. We will use this correspondence later (see Lemma 5.7 (Lemma 7.12).

Recall from Exercise 1.13 i) that $\mathfrak{q} \subseteq r(\mathfrak{q})$. If $\mathfrak{q}$ **is primary** and $\mathfrak{p} = r(\mathfrak{q})$, we say that $\mathfrak{q}$ is **$\mathfrak{p}$-primary**.

Proposition 5.3

Let $\mathfrak{q}$ be a prime ideal of ring A. If $r(\mathfrak{q})$ is maximal, then $\mathfrak{q}$ is primary.

Exercise.

Definition 5.4. A **primary decomposition** of an ideal $\mathfrak{q}$ in A is an expression of $\mathfrak{q}$ as a finite intersection of primary ideals. That is,

$$\mathfrak{q} = \bigcap_{i=1}^{r} \mathfrak{q}_i.$$

Furthermore, we say a primary decomposition is **minimal**, **irredundant**, **reduced**, or **normal** is the followings hold:

1. The $r(\mathfrak{q}_i)$ are all distinct and
2. $\mathfrak{q}_i \supsetneq \bigcap_{j\neq i} \mathfrak{q}_j$ for $(1 \leq i \leq n)$.

§5.2 Primary Decomposition in Noetherian Rings (7.2)

Unless stated otherwise, assume ring A is Noetherian. Let us start with a definition.

Definition 5.5. An ideal $\mathfrak{q}$ is irreducible[1] if

$$\text{(for every ideals } \mathfrak{a} \text{ and } \mathfrak{b} \text{ such that) } \mathfrak{q} = \mathfrak{a} \cap \mathfrak{b} \implies (\mathfrak{q} = \mathfrak{a} \text{ or } \mathfrak{q} = \mathfrak{b}).$$

Remark — From the definition, we say that an ideal $\mathfrak{q}$ is reducible if

$$\text{(there exist ideals } \mathfrak{a} \text{ and } \mathfrak{b} \text{ such that) } \mathfrak{q} = \mathfrak{a} \cap \mathfrak{b} \implies (\mathfrak{q} \neq \mathfrak{a} \text{ and } \mathfrak{q} \neq \mathfrak{b}).$$

Lemma 5.6 (7.11)
Every ideal is a finite intersection of irreducible ideals.

Suppose not. Then, the set

$$S = \{\mathfrak{q} \subseteq A \mid \mathfrak{q} \text{ cannot be expressed as finite intersection of irreducible ideals}\}$$

is non empty. Order S by inclusion. Since A is Noetherian, then the chain

$$\mathfrak{q}_1 \subset \mathfrak{q}_2 \subset q_3 \subset \dots \qquad (\mathfrak{q}_i \in S)$$

terminates somewhere. That is, there exists a maximal element $\mathfrak{q} \in S$. Since $\mathfrak{q}$ is redducible, we have $\mathfrak{q} = \mathfrak{a} \cap \mathfrak{b}$ where $\mathfrak{q} \subset \mathfrak{a}$ and $\mathfrak{q} \subset \mathfrak{b}$. Then, by the maximality of $\mathfrak{q}$, both $\mathfrak{a}$ and $\mathfrak{b}$ are no longer elements of S. This means $\mathfrak{a}$ and $\mathfrak{b}$ can be expressed as a finite intersection of irreducible ideals. Therefore, $\mathfrak{q}$ can also be expressed as a finite intersection of irreducible ideals. A contradiction.

Lemma 5.7 (7.12)
Every irreducible ideal in A is primary.

Proof. Take an irreducible ideal $\mathfrak{a}$ in A. The outline of the proof is as the following:

$$\text{every zero-divisor in } A \text{ is nilpotent } \Rightarrow (0) \text{ is primary in } A/\mathfrak{a} \Rightarrow \mathfrak{a} \text{ is primary in } A.$$

[1] 既約

Now, take non-zero $x, y \in A$ such that $xy = 0$. We show that it is nilpotent. Consider the ascending chain of ideals

$$\mathrm{Ann}(x) \subseteq \mathrm{Ann}(x^2) \subseteq \cdots .$$

This chain terminates somewhere. That is, there exists a positive integer n such that $\mathrm{Ann}(x^n) = \mathrm{Ann}(x^{n+1})$. Now, we claim that

$$(x^n) \cap (y) = (0).$$

□

The two lemmas above show that every ideal other than (1) in a Noetherian ring has a primary decomposition:

Theorem 5.8 (7.13)

Every ideal in A has a primary decomposition.

Proof. Let $\mathfrak{a}$ be an ideal in A. Then:

$$\mathfrak{a} = \underbrace{\bigcap_{i=1}^{r}}_{\text{Lemma 7.11}} \overbrace{\mathfrak{q}_i}^{\text{primary by Lemma 7.12}} .$$

□

Proposition 5.9 (7.14)

Every ideal $\mathfrak{a}$ in A contains a power of its radical.

Let $x_1, \ldots, x_k$ generate $r(\mathfrak{a})$. Let

$$m = \sum_{i=1}^{k} (n_i - 1) + 1,$$

where n_i satisfies $x_i^{n_i} \in \mathfrak{a}$ for each $1 \leq i \leq k$. Then, $r(\mathfrak{a})^m$ is generated by the products $x_1^{r_1} \ldots x_k^{r_k}$ with $\sum_{i=1}^{k} r_i = m$. From the definition of m, we must have $r_1 \geq m_i$ for at least one index i. Therefore, each such monomial lies in $\mathfrak{a}$, and hence $r(\mathfrak{a})^m \subseteq \mathfrak{a}$.

Corollary 5.10

The nilradical in a Noetherian ring is nilpotent.

Proof. Take $\mathfrak{a} = (0)$ From Proposition 7.14. □

Corollary 5.11 (7.16)

Let $\mathfrak{m}$ be a maximal ideal of A, $\mathfrak{q}$ be any ideal of A. Then, the following are equivalent:

i) $\mathfrak{q}$ is $\mathfrak{m}$-primary

ii) $r(\mathfrak{q}) = \mathfrak{m}$

iii) $\mathfrak{m}^n \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ for some positive integer n.

Proof.

i) ⇒ ii) Clear from the definition of $\mathfrak{m}$-primary.

ii) ⇒ iii) Consider

$$r(\mathfrak{q})^n \underbrace{\subseteq}_{\text{Prop 7.14}} \mathfrak{q} \underbrace{\subseteq}_{\text{Ex 1.13 i)}} \mathfrak{m}.$$

iii) ⇒ ii) Consider

$$\mathfrak{m} = r(\mathfrak{m}^n) \subseteq r(\mathfrak{q}) \subseteq r(\mathfrak{m}) = \mathfrak{m}.$$

ii) ⇒ i) Since $r(\mathfrak{q})$ is maximal, by Proposition 4.2 we have $\mathfrak{q}$ is $\mathfrak{m}$-primary.

□

Glossary

English	Japanese	Indonesian
Algebraic terms		
Ring	環	ring / gelanggang
Commutative ring	可換環	gelanggang komutatif
Element	元	elemen / unsur
Set	集合	himpunan
Ideal	イデアル	ideal
Prime ideal	素イデアル	ideal prima
Maximal ideal	極大イデアル	ideal maksimal
Unit	単元	unit
Unit ideal	単位イデアル	ideal satuan
Nilpotent element	冪零元	elemen nilpoten
Nilradical	冪零根基	nilradikal
Jacobson radical	ジャコブソン根基	radikal Jacobson
Radical	根基	radikal
Artinian ring	アルティン環	gelanggang Artinian
Noetherian ring	ネーター環	gelanggang Noetherian
Quotient ring	商環	gelanggang hasil bagi
Direct product	直積	hasil kali langsung
Projection	射影	proyeksi
Homomorphism	準同型	homomorfisme
Ring homomorphism	環準同型	homomorfisme ring
Operations on ideals		
Sum of ideals	イデアルの和	jumlah ideal
Intersection	交わり / 共通部分	irisan
Product of ideals	イデアルの積	hasil kali ideal
Power of an ideal	イデアルの冪	pangkat ideal
Generated by	生成される	dibangkitkan oleh
Family	集合族	keluarga himpunan
Finite family	有限族	keluarga hingga
Only finitely many	有限個だけ	hanya hingga banyaknya
Coprime ideals	互いに素なイデアル	ideal-ideal koprima
Comaximal ideals	互いに素なイデアル	ideal-ideal komaksimal
Greatest common divisor	最大公約数	faktor persekutuan terbesar
Least common multiple	最小公倍数	kelipatan persekutuan terkecil
Componentwise	成分ごと	per komponen
Maps, modules, and exact sequences		
Isomorphism	同型	isomorfisme
Embedding	埋め込み	penyematan
Inclusion map	包含写像	pemetaan inklusi

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English	Japanese	Indonesian
Natural map	自然な写像	pemetaan alami
Injective	単射	injektif
Surjective	全射	surjektif
Kernel	核	kernel
Image	像	citra
Cokernel	余核	kokernel
Restriction	制限	restriksi / pembatasan
Induced map	誘導写像	pemetaan terinduksi
Module	加群	modul
Submodule	部分加群	submodul
Sequence	列	barisan
Exact sequence	完全列	barisan eksak
Short exact sequence	短完全列	barisan eksak pendek
Boundary homomorphism	境界準同型写像	homomorfisme batas
Additive function	加法的関数	fungsi aditif
Flat module	平坦加群	modul datar
Localization and valuation rings		
Integral domain	整域	domain integral
Field	体	lapangan
Field of fractions	分数体	lapangan pecahan
Subring	部分環	subgelanggang
Local ring	局所環	gelanggang lokal
Localization	局所化	lokalisasi
Local property	局所的性質	sifat lokal
Valuation ring	付値環	gelanggang valuasi
Proper ideal	真のイデアル	ideal sejati
Extension of an ideal	イデアルの拡大	perluasan ideal
Integral over	上で整である	integral atas
Integrally closed	整閉	tertutup secara integral
Integral closure	整閉包	penutupan integral
Polynomials and field extensions		
Polynomial	多項式	polinom
Non-constant polynomial	定数でない多項式	polinom tak konstan
Coefficient	係数	koefisien
Exponent	指数	eksponen / pangkat
Root	根	akar
Minimal polynomial	最小多項式	polinom minimal
Algebraic over	上で代数的である	aljabar atas
Transcendental	超越的	transendental
Algebraically closed field	代数的閉体	lapangan tertutup secara aljabar
Algebraic closure	代数的閉包	penutupan aljabar
Finite algebraic extension	有限次代数拡大	perluasan aljabar berderajat hingga
Finitely generated	有限生成	dibangkitkan secara hingga
Generator	生成元	pembangkit

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English	Japanese	Indonesian
k-algebra	k-代数	aljabar-k
Order-theoretic terms		
Partially ordered set	半順序集合 / 順序集合	himpunan terurut parsial
Poset	順序集合	poset
Chain	鎖	rantai
Upper bound	上界	batas atas
Maximal element	極大元	elemen maksimal
Zorn's Lemma	ツォルンの補題	Lemma Zorn
Proof-writing vocabulary		
Suppose	仮に / とする	misalkan
There exists	存在する	terdapat / ada
For any	任意の	untuk setiap
Arbitrary	任意の	sembarang
In particular	特に	khususnya
On the other hand	一方	di sisi lain
Consequently	よって / その結果	akibatnya / maka
Thus	したがって	jadi / maka
Therefore	したがって	oleh karena itu
If and only if	同値である	jika dan hanya jika
Assume for contradiction	背理法で仮定する	andaikan untuk kontradiksi
Contradiction	矛盾	kontradiksi
It follows that	従う	diperoleh bahwa
We denote	と表す / と書く	kita nyatakan dengan
We omit the proof	証明を省略する	bukti dihilangkan
This concludes the proof	これで証明が終わる	ini menyelesaikan bukti
Recall	復習する / 思い出す	ingat kembali
Verify	確かめる / 確認する	verifikasi / periksa
Assertion	主張	pernyataan
Claim	主張	klaim
Converse	逆	konvers
Induction	帰納法	induksi
Without loss of generality (WLOG)	一般性を失わずに	tanpa mengurangi keumuman
Nonempty	空でない	tak kosong
Well-defined	矛盾なく定義される	terdefinisi dengan baik
Extend	拡張する	memperluas
Obtain	得る	memperoleh
Similarly	同様に	serupa / demikian pula
Furthermore	さらに	selain itu
Objective	目標	tujuan
Preparation	準備	persiapan